Session 1: Foundations of Marine Macroecology and Niche Theory

Modeling the distribution of biodiversity under global climate change

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Macroecology and global biodiversity patterns

Macroecology: the big picture

A statistical approach **focusing on large-scale ecological patterns**
(not a separate field of Ecology; coined >30 years ago).

Explores **relationships between biodiversity patterns** (e.g., species abundance levels, and the distribution of genes, species, ecosystems) **and the environment**.

Seeks **the principles governing ecological systems**, ignoring fine-scaled, localized details.

e.g.,

What environmental variables drive the distribution of a mangrove species?

How global climate change may impact kelp forests biodiversity?

Macroecology: the big picture

Compares different **populations** (i.e., different locations), **species** and **ecosystems**, across large areas and long-time periods, **to generalize and scale up results** - ultimately to figure out the global distribution drivers and patterns of biodiversity.

Contrasts with localized, reductionist experimental approaches that may struggle to generate broad predictive theory.

Global Biodiversity Patterns: the non-randomness of life

Species aren't found everywhere – their distributions are limited.

Macroecology helps understand these limits, which are mostly set by:

Niche requirements: e.g., environmental tolerances and resource needs.

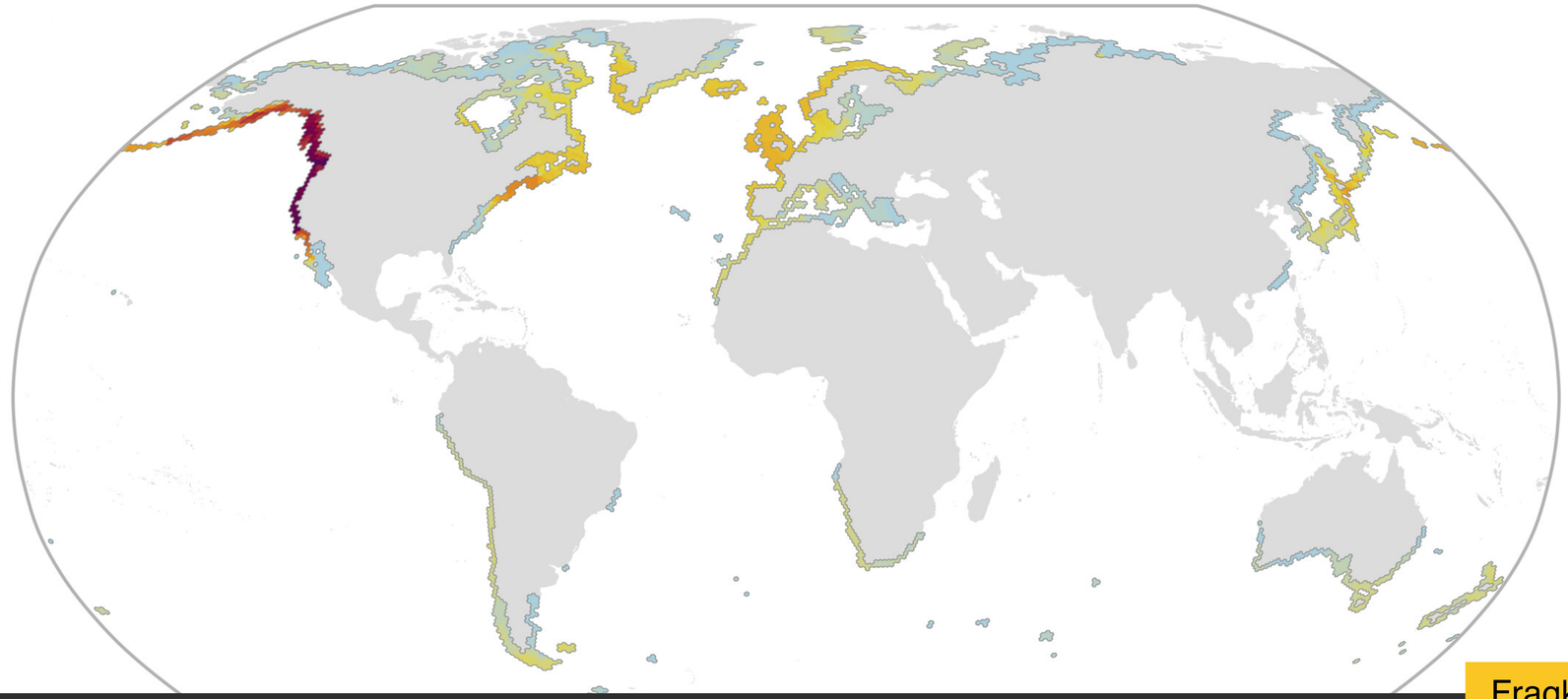
Dispersal constraints: e.g., the ability to reach and establish populations in suitable areas.

Without these constraints, species could potentially occur everywhere, leading to absent or random spatial patterns in distribution and diversity.

Species distributions shape well defined biodiversity hotspots

Multiple **estimates of global species richness and endemicity** for different groups identify **regions with higher and unique biodiversity**.

e.g., *the NE Pacific biodiversity hotspot of marine forests of macroalgae (> 50 species locally).*

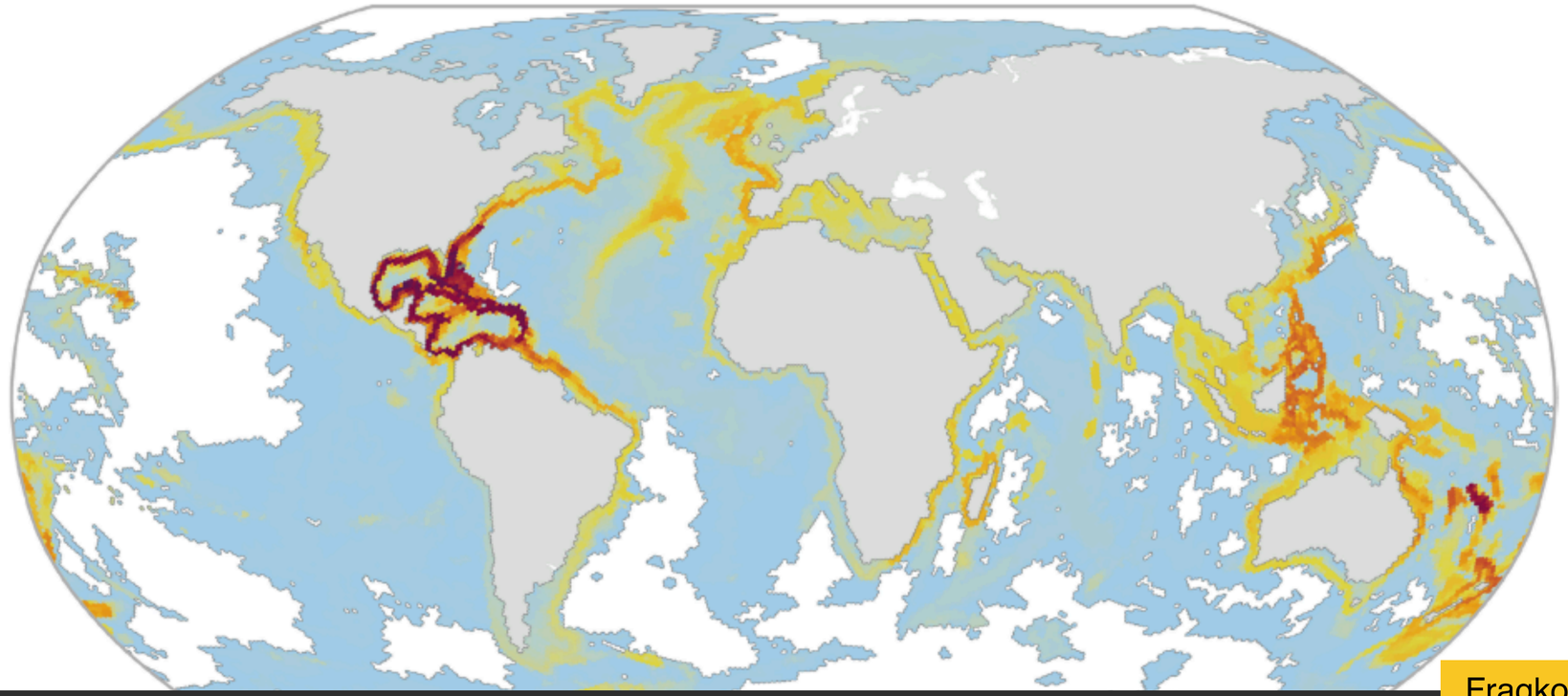


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Species distributions shape well defined biodiversity hotspots

Multiple **estimates of global species richness and endemicity** for different groups identify **regions with higher and unique biodiversity**.

e.g., *the Caribbean Sea and the Indo-Pacific biodiversity hotspot of corals (> 100 species locally).*

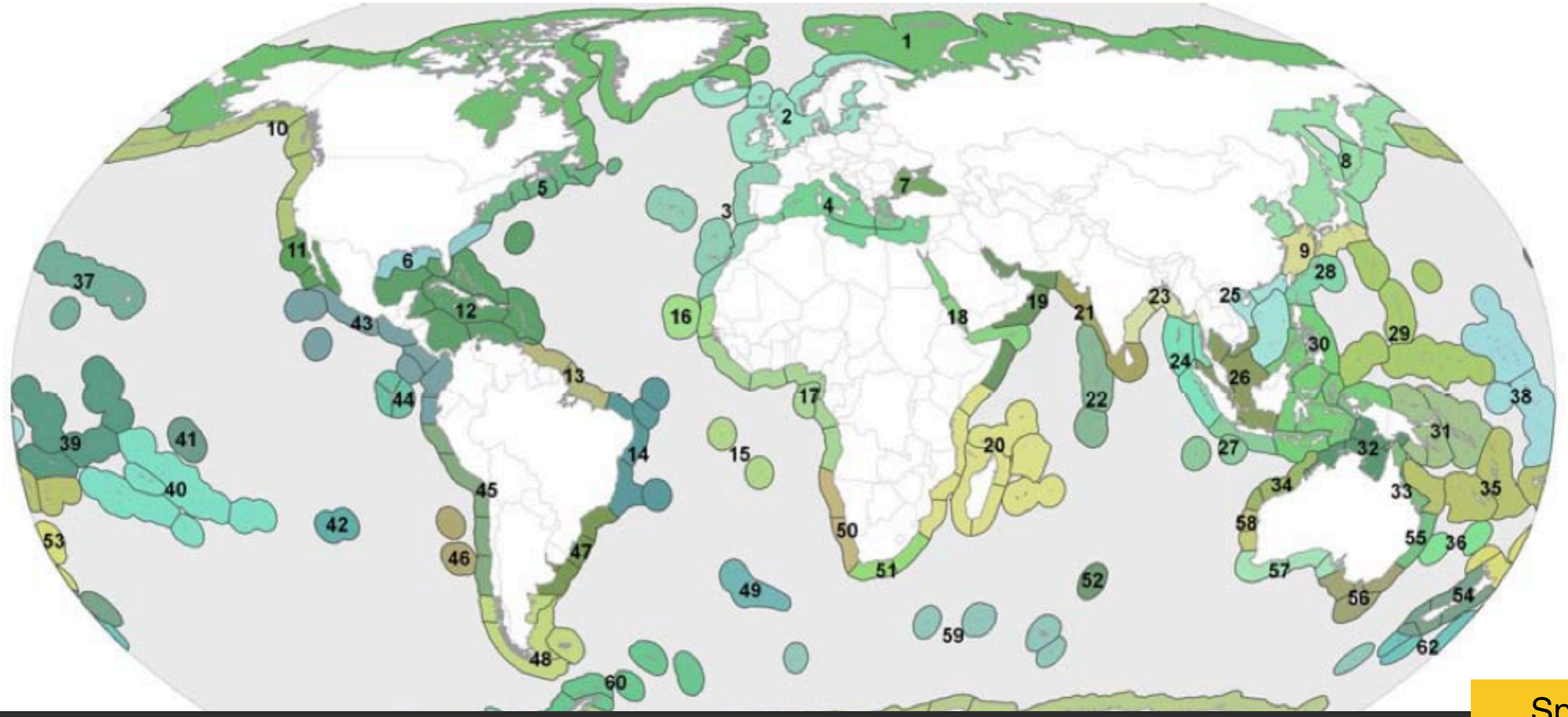


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Species distributions shape well defined biogeographic patterns

Classification systems based on review of published works, historical biogeographic divisions and gradients of abiotic conditions show sharp biogeographic regions.

e.g., *Marine Ecoregions of the World; nested system (12 realms, 62 provinces, and 232 ecoregions).*

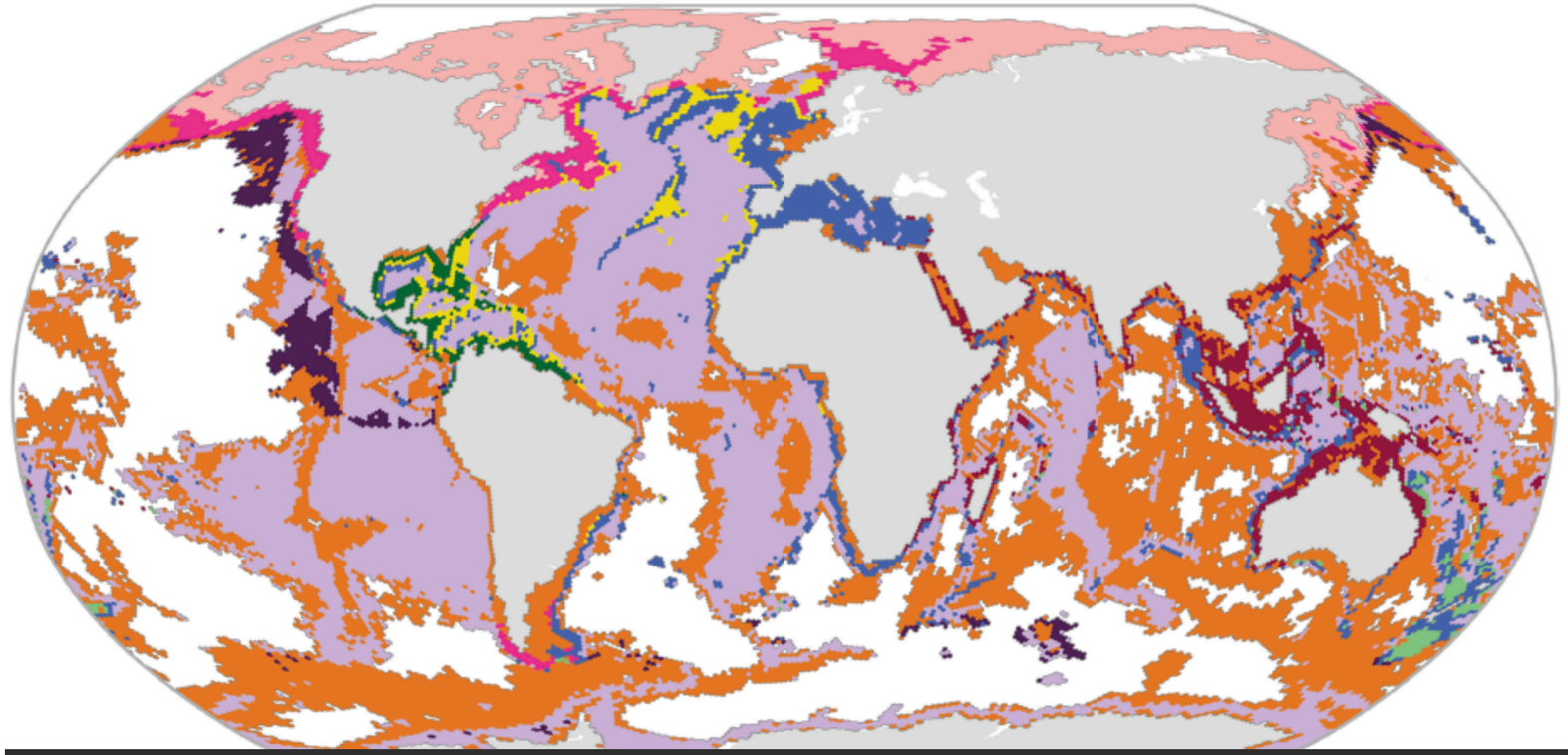


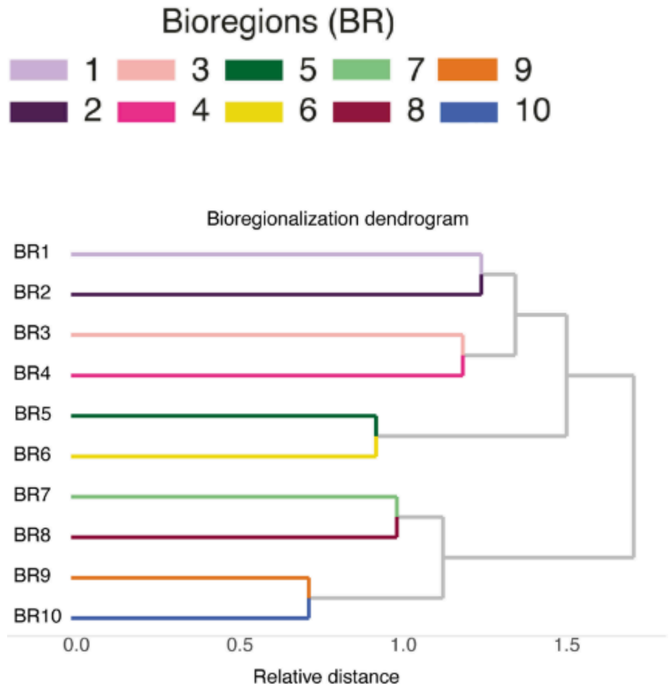
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Spalding et al., 2007

Species distributions shape well defined biogeographic patterns

Classification systems based on **species assemblages** show sharp coral biogeographic regions - within regions similar biodiversity / between regions dissimilar biodiversity.





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